

Project Work: Knowledge Engineering

In this project, your task is to develop a system for detecting fraud in credit card transactions. The goal is to apply the principles of knowledge representation and reasoning.

Organizational Information

- You may work on the project in a team of 2 or alone
- Briefly document your solution in a PDF document: describe the decisions you made during the problem-solving process. Include screenshots or listings so explain the solutions
- Each of the parts 1 to 5 should be one chapter
- In addition to the report, you must provide all files required to build the knowledge base.
- Submit as a ZIP-file named according to the following template:
“<lastname1> <lastname2*> KE-Project.zip”
- The first submission deadline is **15th of June 2026**

Scenario

Every payment made with a credit (or debit) card is stored as a transaction in the billing system of the bank that issued the credit card. A transaction contains and refers to a lot of data. In this project, we want to focus on the following data:

- the payment amount
- the cardholder; here, we are only interested in
 - the name and
 - the country where the cardholder resides
- the merchant; here we are interested in
 - the name,
 - the country where the merchant is based, and
 - the type of service we used (online store, physical store)
- the date and time of the payment.
 - For representing the time you can use a time stamp, which is a number, which gives the time in seconds starting with 0 on 1st of January 1970. To give you an idea: The timestamp of noon on 4th of June 2026 is 1,780,574,400

Rules

The billing system is supported by a knowledge-based system designed to detect fraudulent transactions. It operates on a risk-based approach: it calculates a risk score for each transaction.

To calculate the risk score of a transaction, the following rules are initially applied in this project:

- 1) If the credit card is used in Africa, the risk score increases by 60 points. In Europe, the risk score is 20. For Italy, it is 10.
- 2) If the cardholder is based in Italy, the risk score is not increased. However, if the cardholder is based in another European country, the risk score is increased by 10 points. An African user is assigned 20 risk points.
- 3) If a Swiss person uses the credit card in South Africa, the risk value immediately increases by 30 points. In all other cases, the risk of a "merchant-cardholder restriction" does not increase. It is therefore 0.
- 4) The amount also plays a role in determining the risk score. If the amount is
 - less than €1,000, the risk score is increased by 10 points;
 - up to €10,000, the risk score is increased by 60 points,
 - above €10,000, the risk score is increased by 90 points.
- 5) If the credit card is used in a physical store in two different countries within a 5-minute time difference, then the risk value is at least 150.
- 6) A transaction is stopped, if the risk score is more than 100 points and declined if it is more than 150 points. Declined transactions are called fraudulent transactions.

Test Cases

You can test with the following scenarios:

- A transaction by Mr. Mensoni from Italy, who purchases an expensive wooden mask in a physical store called "World of Africa" in South Africa for €11,234.
- Mr. Meyer from Germany, who pays for a meal worth €12 at the restaurant "Winds of the Deserts" in Morocco.
- Mr Suter from Switzerland pays a meal in the restaurant "Mama Africa" in South Africa for € 90.
- Mr. Mensoni's credit card is used 3 minutes after the purchase of the wooden mask at the restaurant "Winds of the Deserts" in Morocco for a meal costing 500 CHF.



Part 1: Decision Modelling with DMN

- a) Create a DMN model that detects fraudulent transactions. Create a decision model that contains sub-decisions, i.e., a decision whose result is used as input for the main decision.
- b) Is it possible to represent rule 5) in DMN? If you have a solution, create and explain it. If you believe there is no solution, explain why

Part 2: Ontology

This project uses a terminology that includes the most important terms and the relationships between them. Create a graphical ontology that defines the concepts and relationships as properties. Data about transactions must **not** be in the graphical ontology.

Part 3: Prolog

- a) Create a knowledge base in Prolog that detects fraudulent transactions. Consider what information about the transaction and its stakeholders must be represented and what are suitable predicates to represent the information.
- b) Represent the test cases in Prolog
- c) Formulate appropriate queries.

Part 4: Knowledge Graph and Rules

- a) Define an RDF schema for the objects and properties relevant to the detection of fraudulent transactions.
- b) Represent the test cases in RDF.
- c) Use SPARQL to retrieve
 - all transactions
 - transactions with merchants from South Africa
 - transactions with cardholders from Europe
 - transactions and their merchant risk for transactions with merchant from Italy
- d) Create SWRL rules that derive for each transaction whether it is fraudulent. Then use SPARQL to retrieve the fraudulent transactions.



Part 5: A-B Transaction

The terminology has been updated: We distinguish between "A-transaction" and "B-transaction" which are subclasses of "transaction."

- Every transaction made in an online store is classified as an A-transaction (instance of the class "A-transaction"). Currently these are the only rules for A transaction which we know. Additional rules might be added if required.
- A "B-transaction" is the opposite of an "A-transaction", i.e. every action, which is not an "A-transaction" is a "B-transaction"

You can test with the following transactions:

- Mr. Miller from Germany orders something for € 102 from the online store "South African Master Pieces" in South Africa.
- Mr Suter from Switzerland pays a meal in the restaurant "Mama Africa" in South Africa for € 90.

- a) Represent the rules for "A-transaction" and "B-transaction" in Prolog.
- b) Is it possible to create SWRL rules to identify "A-transaction" and "B-transaction"?
- c) Can A and B transactions be retrieved with SPARQL? If not, why not?

